

Sky-Recovery

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API Documentation

The Sky-Recovery backend exposes a RESTful API serving the React frontend. It handles passenger authentication, data retrieval, recovery logic processing, and generative AI chat integration.

Base URL

All API endpoints are prefixed with `/api` unless otherwise specified.

Authentication (`/api/auth`)

`POST /auth/login`

Authenticates a passenger using email and password.

- **Request Body:** `{ email, password }`
- **Response:** Passenger details including `id`, `firstName`, `lastName`, `loyaltyStatus`, etc.

`POST /auth/forgot-password`

Initiates a password reset flow (simulated).

- **Request Body:** `{ email }`
- **Response:** Reset token (for demo purposes)

`POST /auth/reset-password`

Completes the password reset process.

- **Request Body:** { token, newPassword }
 - **Response:** Success confirmation
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Passengers (/api)

GET /passenger/:id

Retrieves a passenger by their UUID, including their bookings and flight details.

- **Response:** Detailed Passenger object

GET /passenger/email/:email

Retrieves a passenger by their email address.

- **Response:** Detailed Passenger object

GET /passengers

Retrieves all passengers (utility endpoint for demo).

Flights & Bookings (/api)

GET /flights

Retrieves all flights in the system.

GET /flights/:flightId

Retrieves a specific flight and all active disruptions associated with it.

Disruption Recovery (/api/recovery)

POST /recovery/analyze

Core engine endpoint. Analyzes a disrupted booking and generates AI-driven recovery recommendations using the Gemini API based on loyalty tier, flight schedules, and delay reasons.

- **Request Body:** { bookingId }
- **Response:** A proposed RecoveryRecommendation (Rebook, Refund, or Wait)

POST /recovery/accept

Accepts a proposed recovery recommendation (e.g. Reschedule).

- **Request Body:** { bookingId, recommendationId, selectedFlightId (optional) }
- **Response:** Updates the request status to PENDING (simulating admin review).

POST /recovery/refund

Requests a refund for a disrupted flight.

- **Request Body:** { bookingId }
- **Response:** Updates the request status to PENDING (simulating admin review).

GET /recovery/history/:bookingId

Fetches the historical recovery requests for a specific booking.

Generative AI Chat (/api/chat)

POST /chat

Streams or returns a generated response from Google's Gemini AI acting as the SkyRecover Assistant.

- **Request Body:** { passengerId, message }
- **Response:** AI Assistant reply based on live passenger context (e.g. "Your flight SJ238 is delayed due to weather").

GET /chat/:passengerId

Retrieves the chat history between the passenger and the Gemini AI.

Notifications (/api/notifications)

GET /notifications/:passengerId

Fetches all unread notifications for a passenger.

POST /notifications

Pushes a new notification to a passenger.

- **Request Body:** { passengerId, type, title, description }

POST /notifications/:passengerId/read-all

Marks all notifications as read for a passenger.

Evaluation / Demo Utilities (/api/demo)

These endpoints exist purely for the hackathon evaluation to simulate live operational events.

POST /demo/delay-flight

Simulates a flight delay and injects a disruption record into the database.

- **Request Body:** { flightId, delayMinutes, reason }

POST /demo/cancel-flight

Simulates a flight cancellation.

- **Request Body:** { flightId, reason }

POST /demo/reset

Resets the database for a specific user to its original seeded state to allow judges to re-evaluate the recovery flow.

- **Request Body:** { email }